

A Review of Cardiac MRI Reconstruction Techniques

ET4399 Research Project at University
Delft University of Technology

Burcu Erarslanoglu (5948355)

Abstract

Cardiac Magnetic Resonance Imaging (CMRI) is an extremely useful tool for assessing cardiovascular function due to its high spatial resolution and non-invasive nature. However, due to cardiac and respiratory motion, CMRI comes with the challenges of motion artifacts, blurring, and ghosting. To address these challenges, synchronization with the cardiac and respiratory cycles is often adopted but those alone can not be enough when clinical feasibility is considered. This review explored traditional CMRI acquisition and reconstruction methods including spin echo and gradient echo-based techniques, parallel imaging, and compressed sensing. Additionally, recent advancements in image reconstruction using deep learning, hybrid approaches, and low-rank plus sparse matrix decomposition are discussed for their potential to enhance speed and image quality for CMRI. In a comparative analysis, the strengths and limitations of these techniques are highlighted. Despite the progress achieved, challenges such as computational demands and integration into clinical workflow need to take place. Future advancements in MR technology and data-driven approaches hold the potential to overcome these barriers and open the way for more efficient and accurate cardiac MR images.

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1 Introduction

Magnetic Resonance Imaging (MRI) is widely used in medical fields to generate detailed images of internal body structures. The high-resolution MR images aid in the diagnosis and treatment of various conditions [1]. Cardiac MRI (CMRI) extends these imaging capabilities to the heart by capturing dynamic and high-resolution visuals which are essential for understanding cardiovascular function. CMRI is a reference standard in cardiology in clinical and research settings because of its versatility, high reproducibility, and accuracy [2]. It has proved its superiority over other established non-invasive modalities in terms of assessment of cardiac and great vessel morphology and left ventricular function [3]. The advantages of CMRI over other modalities such as Echocardiography or Cardiac Computed Tomography (CT) are due to its high spatial resolution, tissue characterization abilities, and lack of ionizing radiation [4].

While standard MRI provides detailed static images of body structures, CMRI is intended for the heart. To assess cardiac structure and function, there are several modalities of CMRI with different properties and applications. These include cardiac cine MRI, Coronary MR angiography (MRA), and phase contrast MRI [2]. Among these, cine imaging is the most common technique in CMRI and it is considered as the gold standard for the assessment of cardiac function [5]. Cardiac cine MRI captures a series of images of the beating heart, which are then combined to create real-time visuals of the heart in motion. The heart is usually imaged at a single slice location throughout the cardiac cycle, including 10 to 30 cardiac phases. This dynamic technique allows clinicians to observe cardiac function across multiple phases of the cardiac cycle [6]. In other words, by producing a series of images of the heart, cardiac cine MRI creates a “movie” of heart activities.

Cine CMRI can give a complete cardiovascular assessment in a single setting; however, it is considered as a technically challenging imaging modality. With ordinary MRI, the generation of high-resolution still images of internal organs is possible, as mentioned before. However, when a moving organ such as a beating heart needs to be imaged, special techniques are required to capture clear and accurate images [7]. The challenge stems from accelerating the inherently slow data acquisition process while maintaining high resolution and image quality. Due to the slow image acquisition, imaging moving organs comes with the undesired effects of image blurring and ghosting [2]. The motion of the heart causes parts of the organ to appear smeared, resulting in blurring. When the blurring artifact occurs, the images are not sharp, and fine details can not be distinguished. On the other hand, ghosting creates faint and repeated images of the moving structures because the motion disrupts the steady acquisition of the data.

Blurring and ghosting artifacts are especially prevalent in CMRI due to respiratory motion and the continuous beating of the heart. They obscure critical information about the cardiac anatomy and function, inhibiting correct assessment and diagnosis. To address these challenges, techniques like cardiac gating and respiratory gating [8] could synchronize MR data acquisition to the cardiac or respiratory cycles. Cardiac gating synchronizes image acquisition with the heart’s electrical signals, as measured by an electrocardiogram (EKG). The cardiac image is produced in one R-R interval, in other words in between two heartbeats [1]. In respiratory gating, the patient’s breathing cycle is monitored by using sensors and the image acquisition of the MRI takes place only in certain parts of the breathing cycle when the chest is mostly still. In practice, cardiac and respiratory gating can be combined, for example with the dual cardiac-respiratory gating device [9], to reduce the image artifacts due to cardiac and respiratory motion even further. Breath-holding [10] is another technique used to reduce motion blur where the patient is asked to hold their breath for the duration of the image acquisition. However, this simpler solution to the challenge of artifacts may not work with most patients who have difficulty holding their breath such as those with respiratory conditions. Additionally, breath-holding may not be feasible for longer imaging

sequences which may take up to 45 to 90 minutes, where the breath-hold duration becomes multiple of 10 to 20 seconds during the full scan time [11]. This is an unrealistic expectation, especially for those undergoing cardiac diagnosis [12].

CMRI studies employ a range of different acquisition and reconstruction methods to improve image quality and reduce motion artifacts related to the cardiac or respiratory cycles. The common acquisition techniques of breath-holding, cardiac gating, and respiratory gating often come with some limitations. These methods may not be practical when the breath-hold requirements are considered. Also, the need for additional instruments for the gating methods makes them time-consuming and resource-intensive. To tackle these limitations, advanced reconstruction methods of parallel imaging [1] and compressed sensing [13] are widely used. These reconstruction methods aim to accelerate data acquisition while maintaining a high-resolution. However, even these methods have certain limitations. For that reason, the exploration of machine learning (ML) techniques particularly deep learning (DL) is required. These techniques can improve the imaging process by using algorithms that learn from vast amounts of imaging data. Thus, with them reconstruction quality can be improved and the speed of MRI scans can be fastened [14].

The objective of this paper is to present and compare traditional and emerging CMRI reconstruction techniques. Their strengths, limitations, and challenges that remain in achieving high-resolution images for accurate cardiac assessment will be discussed. In this paper, the traditional CMRI acquisition and reconstruction techniques are first presented in section 2. Later, the cutting-edge techniques that mostly emerged in the past decade are explained in section 3. The presented methods are compared from multiple aspects in section 4. The challenges and limitations that remain with the current state-of-the-art are elaborated on in section 5. The paper concludes with future research directions that could be taken to address the remaining challenges and a summary of the key findings in section 5.

2 Traditional Cardiac MRI Acquisition and Reconstruction Techniques

Traditional cardiac MRI techniques encompass a range of hardware optimizations and algorithmic approaches. They include both acquisition sequences, which determine how MR data is collected; and reconstruction techniques, which process the acquired data into interpretable images. Methods based on spin echo (SE) sequences and gradient echo (GRE) are mostly concerned with optimizing the MRI hardware and improving the physics of signal acquisition [1]. Although these methods primarily refer to the acquisition process, they influence how the images are later reconstructed especially in cardiac MRI. Parallel imaging uses multiple receiver coils to acquire spatial data simultaneously to accelerate image acquisition [1]. Additionally, compressed sensing uses advanced algorithms to reduce the acquisition times by reconstructing high-quality images from undersampled data [15]. These methods are designed to improve image quality, reduce artifacts, and optimize scanning efficiency in MR imaging and cardiac imaging.

2.1 Spin Echo-Based and Gradient Echo-Based Methods

GRE sequences provide faster image acquisition than SE sequences because they use magnetic gradients to generate a signal, instead of applying two radio-frequency (RF) pulses [1]. Despite these differences, both sequences have been extensively applied over the years for CMRI. This part of the literature review will focus on two of the GRE-based methods, namely FLASH (Fast Low Angle Shot) and SSPF (Steady-State Free Precession); and a SE-based method TSE (Turbo Spin Echo). These methods were chosen due to their high relevance to the context of cardiac imaging, and will be explained briefly according to their chronological development.

2.1.1 FLASH (Fast Low Angle Shot)

FLASH was invented in 1985 by Haase and colleagues [16]. It introduced gradient echo (GRE) sequences with low flip angles which enabled faster imaging compared to conventional spin echo (SE) techniques. The technique provided abdominal images without respiratory motion and digestive motion artifacts [17]. FLASH made the dynamic imaging of the heart possible for the first time, but the acquisition process still required synchronization with EKG, usually together with breath holding [10], [18]. The FLASH sequence has many variants that have been successfully applied for CMRI; these include segmented TurboFLASH [10] introduced in 1991 and radial FLASH introduced in the mid-1990s and advanced further till the 2010s [18]. The comparison between different methods will be provided in section 4.

2.1.2 TSE (Turbo Spin Echo)

TSE -also known as FSE (Fast Spin Echo)- is the commercial name for the RARE (Rapid Acquisition with Relaxation Enhancement) technique which was originally proposed in 1986 by Hennig et al. [19]. Different MRI manufacturers often implement similar sequences under various names; therefore, multiple terms for the same sequences can be found in the literature. TSE was a significant advancement because of its ability to reduce scan times by using RF planar principles [20]. Compared to conventional SE techniques, the acquisition times are reduced significantly with satisfactory image qualities and contrasts. The fast nature of TSE makes it appropriate to use in scenarios where fast scans are required; for example for CMRI where breath-hold acquisitions are required. An important variant of the TSE sequence is HASTE (Half-Fourier Acquisition Single-shot Turbo Spin Echo) which is a single-shot technique for very fast imaging with half-Fourier acquisition [21].

2.1.3 SSFP (Steady-State Free Precession)

SSFP refers broadly to MRI techniques that maintain a steady-state of transverse magnetization by using balanced gradient pulses across multiple cycles [1]. The steady-state nature of SSFP allows for rapid imaging with a high signal-to-noise ratio (SNR) and this property makes it suitable for dynamic imaging applications, such as cardiac cine MRI. A variety of different SSFP methods exist, and manufacturing companies have chosen to name and brand these methods differently. Despite the varied terminology, these techniques all operate under the core principles of SSFP. A summary of these techniques can be found in Table 1.

Table 1: Classification of SSFP MRI techniques by manufacturers. [3]

Classification	SSFP		bSSFP
	FID-like	Echo-like	
Siemens	FISP	PSIF	TrueFISP
GE	GRASS	SSFP	FIESTA
Philips	FFE	T ₂ -FFE	b-FFE

The bSSFP stands for “Balanced SSFP” and it uses fully balanced gradients in all spatial directions [1]. Two primary forms of SSFP exist, as can be seen. These are FID-like SSFP and echo-like SSFP. FID-like SSFP emphasizes initial free induction decay (FID) signals, producing images with high temporal resolution. On the other hand, echo-like SSFP capture signals similar to gradient-echo

sequences and are often used for T_2 -weighted imaging. Techniques that use principles of SSFP have been heavily applied to CMRI because of their success. The segmented cine TrueFISP is a good example for cardiac application since it used both SNR and blood-myocardial contrast-to-noise ratio (CNR) to evaluate the performance of the method [22].

2.2 Parallel Imaging

The method of parallel imaging replaces part of the k-space data with spatial information gathered from additional RF coils. In traditional MRI, phase encoding gradients are applied to sample the entire k-space. Parallel imaging, first introduced in the late 1990s, offers an alternative to this time-consuming process by the use of additional RF coil data. This process reduces scan times drastically and allows for faster data acquisition [1], [23]. Since parallel imaging reduces scan times, it is particularly useful for CMRI, where patient comfort and minimizing motion artifacts are crucial. Below two of the commercially used parallel imaging methods are explained. Both approaches achieve faster scan times without compromising image quality. Temporal filtering can be applied to dynamic auto-calibrated imaging.

2.2.1 SENSE (Sensitivity Encoding)

SENSE uses multiple receiver coils to reduce scan time, allowing partial sampling of the k-space by using each coil's unique spatial sensitivities. It enables faster imaging without fully relying on phase encoding gradients. This approach is a key method in parallel imaging, where multiple coils work in parallel to accelerate data acquisition by leveraging their distinct sensitivity profiles. SENSE does not compromise spatial resolution, but it may involve trade-offs, such as increased image noise. This approach has been shown to cut scan times by half with two coils and by up to two-thirds with an array of five coils [24], particularly benefiting applications like brain and cardiac imaging. The feasibility of this approach for cardiac real-time imaging of human subjects was verified in 2000 by Weigner and colleagues [25]. SENSE is commonly used in Philips MRI systems [23].

2.2.2 GRAPPA (Generalized Autocalibrating Partially Parallel Acquisitions)

SENSE requiring the prior information of the coil sensitivity profiles can be a drawback because coil sensitivities change under load and at higher frequencies in practice [1]. GRAPPA reconstructs missing k-space lines by using data from multiple receiver coils, it does so by not relying on a composite signal from all coils. Instead, it individually fits each coil's signal to a single calibration line in k-space, known as the Autocalibration Signal (ACS). With this process, unique linear weights for each coil can be determined to be used to fill in missing k-space data [26]. GRAPPA is widely implemented in Siemens MRI systems [23].

2.3 Compressed Sensing

The data collected from an MRI scanner consists of samples from the spatial Fourier transform of the imaged objects [27]. The Nyquist rate determines the minimum sampling frequency required to reconstruct the MR images accurately. In many applications, such as this one, one must sample at or above the Nyquist rate to ensure proper image reconstruction and avoid aliasing artifacts [28]. However, due to the Nyquist rate and the requirement of having high-resolution and clear images in MRI; the imaging times get very long. This makes MRI a slow imaging modality. Depending on the task, scans can take up to several minutes to hours [12]. This is of course not practical when the patients and resources in healthcare are considered. Compressed sensing is often employed to accelerate the scans. With compressed sensing, a minimum number of samples can be taken with random undersampling by using the underlying sparsity in the MRI data [28].

A compressed sensing problem formulation for MRI was first proposed by Lustig et al. [15] in 2007. This problem formulation is suitable for MRI for two main reasons [13]. Firstly, MR images are inherently compressible through sparse coding in an appropriate transform domain. Sparsifying transforms such as the Discrete Cosine Transform (DCT) or Discrete Wavelet Transform (DWT) can map the image information into a vector. This vector would have a few significant coefficients and this allows using only the most important coefficients while ignoring the smaller ones. Secondly, MR images contain spatially encoded samples that lie in k-space rather than direct pixel samples which enables the reconstruction of high-quality images from a small subset of data. The small subset of data that is used for reconstruction is linked with the concept of incoherent sampling where data is acquired in a random or non-uniform manner [12], [28]. Figure 1 can be viewed to understand the workflow of compressed sensing in MRI.

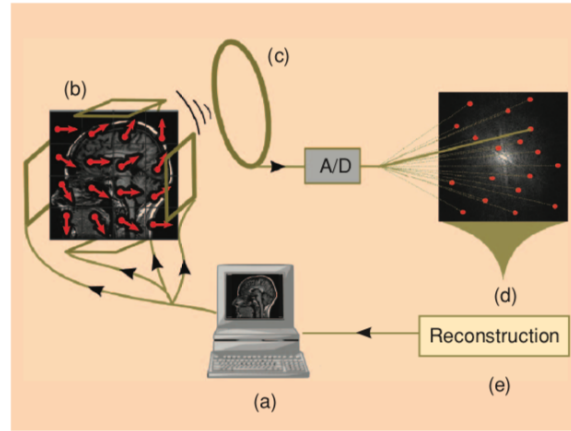


Figure 1: MRI as a compressed sensing system. [13]

Workflow in Figure 1 shows; (a) the gradient waveforms being adjusted by the system to allow incoherent sampling of k-space, which (b) regulates the phase of the image pixels or voxels. (c) The RF coil captures the signal encoded as samples in k-space, which are then digitized by an analog-to-digital converter (ADC). (d) The incoherently sampled k-space data is finally (e) reconstructed into an image using compressed sensing by enforcing sparsity constraints. Overall, the central advantage of compressed sensing in MRI is reducing imaging time while preserving high-resolution image quality. As a result, the diagnostic quality is not compromised.

CMRI can make use of compressed sensing as well and reduce the motion artifacts of blurring and ghosting by allowing a lower scanning time. CMRI has some advantages compared to other methods, such as X-ray coronary angiography or multi-slice X-ray CT, that are used for evaluating the heart, its coronary arteries, and in general coronary artery diseases. Due to cardiac catheterization [29], the method of X-ray coronary angiography is considered to be invasive while CMRI is not. Its variation of multi-slice X-ray CT is non-invasive but it uses high doses of ionizing radiation [30] which is therefore considered relatively less safe when compared to CMRI. Despite these benefits, imaging coronary arteries with CMRI presents challenges due to the constant movement of the coronary arteries. Techniques such as synchronizing imaging with the cardiac cycle or respiratory cycle that were mentioned in section 1 are still applicable and can compensate for the heart-beating motion or respiratory motion. However, often times patients are also required to hold their breath for long times to achieve a still image, which complicates obtaining high-resolution images. The integration of compressed sensing in CMRI can reduce the motion artifacts by allowing shorter scan times, potentially enabling the entire heart to be scanned in a single breath-hold.

3 Recent and Emerging Cardiac MRI Reconstruction Techniques

Significant progress has been made in MRI reconstruction over the recent years. The recent and emerging techniques integrated cutting-edge technologies while going beyond traditional methods. The goal of the latest approaches was mainly related to reducing acquisition time without sacrificing image quality. Most of the recent techniques were able to enhance image quality as well. These methods include a variety of ML and DL techniques that use large datasets and powerful computational models. They reconstruct high-quality MR images by making predictions. Furthermore, there are hybrid techniques that leverage the strengths of multiple techniques. Besides the hybrid techniques, matrix-based techniques such as low-rank plus sparse matrix decomposition provide an alternative framework for MRI reconstruction. It makes use of the low-rank structure and sparsity in MRI data. The methods to be explained in this section represent a transition to data-driven approaches in MRI reconstruction while addressing the limitations of earlier techniques. Although some methods discussed in this section were developed for general MRI applications, they can also be effectively applied to CMRI.

3.1 Deep Learning-Based Techniques

ML is the science of programming computers to enable them to learn from data. DL, which will be the focus of this section, is a subset of ML mimicking the way the human brain operates by forming a neural network [31]. In a neural network, there are three types of layers: input, output, and hidden layers. Each layer contains interconnected nodes that process inputs via the activation functions. These activation functions are often non-linear, allowing the neural network to learn complex patterns in the data. The processed inputs at each layer produce outputs that are passed to the subsequent layers in the network. Artificial Neural Networks (ANNs) are the foundational form of DL; however, specialized types such as Convolutional Neural Networks (CNNs), also exist. CNNs are particularly useful for image processing applications [32] including MR image reconstruction.

For MR image reconstruction, unrolled networks have been heavily studied. The state-of-the-art methods use unrolled networks [33]. These methods use the traditional iterative optimization algorithms in the compressed sensing MRI framework, to obtain an end-to-end neural network. When an unrolled network is used to reconstruct an MR image, the neural network can learn the entire process of mapping between the raw k-space data to a final reconstructed image. In this setup, image transformations, sparsity-promoting functions, regularization parameters, and update rates are used for training purposes while also allowing optimization during training [14], [34]. The optimization algorithm of the iterative shrinkage-thresholding algorithm (ISTA) has been successfully applied by Pezzotti et al. for undersampled knee MRI reconstruction in [35]. This application of adaptive intelligence, to accelerate MR acquisition can be found in subsubsection 3.2.2.

Recent studies and developments show extremely promising results for the application of DL in the cardiac cine MRI reconstruction task because DL can enable faster acquisitions at higher resolutions. In one study by Masutani and colleagues DL was used for single-frame and multi-frame resolution for CMRI [36]. In this study, short-axis cine CMRI scans were used with a CNN for training purposes. The performance with the CNN outperformed traditional methods, demonstrating its potential for enhancing image quality and reducing reconstruction time. Another very important work was the development of CINENet, a DL-based 3D cardiac cine MRI reconstruction method. This method employs complex-valued 4D spatio-temporal convolutions and multi-coil data processing to achieve isotropic resolution in a single breath-hold and reduces both acquisition and reconstruction times dramatically compared to conventional methods [5]. Additionally, artifact reduction techniques for 2D radial cardiac cine MRI have also benefited from DL, as demonstrated in a study utilizing a specialized CNN known as U-Net [37], which was originally developed for

biomedical image segmentation. On top of these developments, there have also been some industry innovations such as GE HealthCare’s Sonic DL™ technology [38]. This technology has further advanced CMRI by using DL to accelerate image acquisition, achieving up to 12 times faster scans. This technology developed in January of 2024 is able to perform MR scans within a single heartbeat so in one R to R. A comparison between an image obtained with Sonic DL™ and conventional SSFP demonstrates the power of this technology in Figure 2 below.

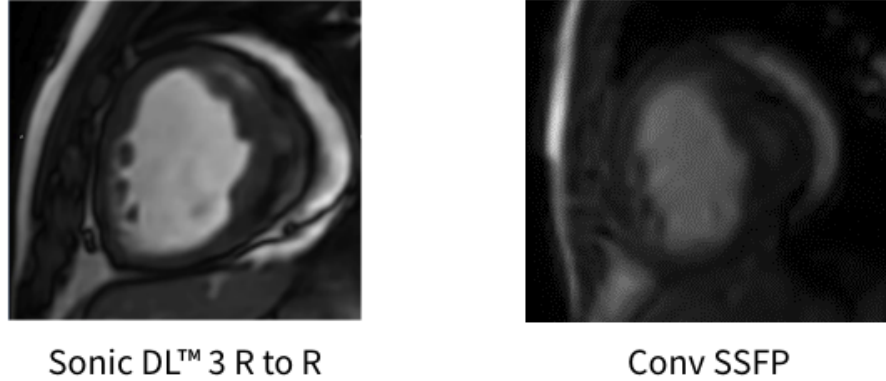


Figure 2: Comparison of cardiac MR images of a 14-year-old patient with congenital heart disease. [38]

The listed advancements not only illustrate the versatility of DL architectures in addressing diverse challenges in cardiac MRI but also emphasize the potential for further innovations.

3.2 Hybrid Techniques

Hybrid MRI techniques combine multiple imaging and reconstruction methods to improve speed and image quality. These approaches often use both compressed sensing (explained in subsection 2.3) and parallel imaging (explained in subsection 2.2) to reduce scan times while also maintaining image quality. Compressed SENSE and Adaptive-CS-Net are two examples of hybrid techniques that integrate traditional MRI methods with advanced algorithms.

3.2.1 Compressed SENSE

Compressed SENSE is a technology developed by Philips in 2017 that involves both the acquisition and reconstruction processes. It integrates the principles of compressed sensing with SENSE (specifically dS SENSE) to achieve faster MRI scans [39]. As mentioned before, compressed sensing samples the k-space incoherently to reduce the scanning times. However, this can result in a lower SNR since the k-space is not fully sampled. Due to sparsity, the scan captures only the most important MR signals while the rest are ignored. The system iteratively reconstructs images from undersampled k-space data using knowledge-based algorithms, filling the empty lines in the k-space. In each iteration, sparse k-space data is transformed into the actual MR image by going through some steps. These are the Fourier transform which introduces noise; and the Wavelet transform, decomposition, and filtering steps to remove the noise and enhance image quality. The improved image is used to find the improved k-space data which is then used to fill out sparse k-space data. Achieving the optimal image might require multiple iterative reconstruction loops. Compressed SENSE can eliminate artifacts and produce high-quality diagnostic images. It supports both 2D

and 3D scans for all anatomies -including the heart- and contrasts [40]. Below is a freeze frame from cardiac cine imaging of the heart showing all four chambers.

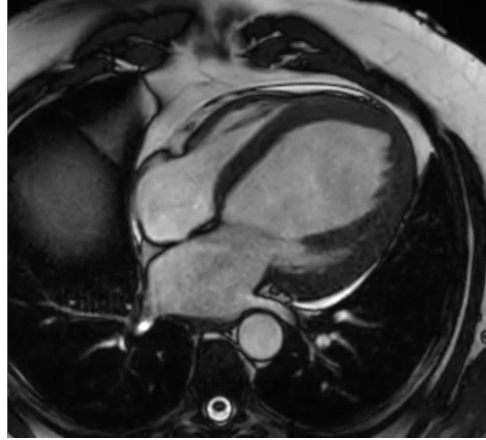


Figure 3: bTFE (balanced Turbo Feild Echo) cine 4 chamber imaging using Compressed SENSE. [41]

3.2.2 Adaptive-CS-Net

As previously mentioned, unrolled networks work successfully for MR image reconstruction. The Adaptive-CS-Net was designed to reconstruct undersampled MRI data, specifically for the 2019 fastMRI challenge [35]. At the core of this approach, there is ISTA and its DL variant of ISTA-Net. The Adaptive-CS-Net approach simulates an iterative compressed sensing reconstruction by unrolling the optimization process into a fixed number of neural network layers. The network incorporated a multiscale sparsification, similar to Wavelet transform which helps to produce more accurate reconstructions. It uses a 2.5D learning approach where the neighboring slides are included as inputs to the network. This allows the network to use spatial correlations across slices without the computational cost of full 3D convolutions. Also, the network uses soft priors based on MRI-specific characteristics so that the network can be guided to learn some features directly. Trained using Compressed SENSE data, the network combines model-based compressed sensing techniques with data-driven (DL) elements, achieving high performance in MRI reconstruction.

Pezzotti and colleagues have shown Adaptive-CS-Net’s success for undersampled knee MRI reconstruction in [35]; however, this method was then further developed to accommodate a wider range of acceleration factors and anatomical features. In 2023, Dratsch et al. worked on the reconstruction of shoulder MRI by using the compressed SENSE acquisition scheme and CS-AI which employed the Adaptive-CS-Net. The shoulder images reconstructed with CS-AI demonstrated improved subjective image quality, supported by enhancements in SNR and CNR metrics.

3.3 Low-Rank Plus Sparse Matrix Decomposition

The origin of the low-rank plus sparse matrix decomposition is the matrix completion techniques that use the structured low-rank framework for the recovery of an image or a series of images from a few measurements [42]. The low-rank plus sparse matrix decomposition aims to separate a dynamic dataset into two components by using the $\mathbf{L} + \mathbf{S}$ model; where $\mathbf{L}$ captures the slowly varying background information and $\mathbf{S}$ captures the dynamic features such as motion or contrast changes. In this model, the low-rank matrix is represented with $\mathbf{L}$, and the sparse matrix is represented with $\mathbf{S}$ [42], [43]. This model is used in practice to describe the dynamic MRI data model because the k-t space data, where “t” represents how the spatial information progresses over time, has a low-rank

and sparse structure as well. Dynamic MRI models and hence the low-rank plus sparse matrix decomposition technique can be used in many different MRI applications. These include cardiac perfusion, cardiac cine, and time-resolved angiography [43]. Another related reconstruction technique is low-rank matrix completion (LRMC), for undersampled magnetic resonance fingerprinting (MRF) data. This method can address different tissue properties [44].

This method can be adopted for the undersampled dynamic MRI reconstruction for multiple significant reasons. The static anatomical structures or the slowly varying signals that do not contribute to the actual anatomical feature being imaged are shared across the temporal frames. These background structures are said to have “strong spatiotemporal correlations” [43] and therefore exhibit a low-rank structure. This makes the data compressible. Therefore, when its singular value decomposition (SVD) is considered, only a few singular values significantly contribute to the data model. This essentially implies that truncating the SVD would give a compressed representation of the entire k - t space data model. As for the sparse component $\mathbf{S}$, the dynamic features often occupy only a small portion of the image space. Their sparse nature means a few non-zero entries in the temporal frequency domain of the data. Overall, the $\mathbf{L} + \mathbf{S}$ decomposition demonstrates significant benefits for dynamic MRI applications including cardiac cine MRI. The method’s ability to separate the background and dynamic components improves both spatial and temporal resolutions while allowing the use of high-acceleration factors [43].

4 Comparative Analysis of Traditional and Emerging Techniques

Cardiac cine MR images are often compared based on speed, subjective and objective image quality, and applicability to clinical domains. With subjective image quality, researchers can assess the overall anatomical features and visualize the artifacts such as ghosting and blurring, the most notable ones for CMRI. With objective image quality, quantitative evaluation metrics are used. In this section, a thorough comparison will be carried out for the traditional and emerging techniques that have been discussed so far.

The traditional methods, especially the SE and GRE-based methods have aimed to optimize speed without compromising image quality. In their paper from 1991, Atkinson and colleagues made a comparison between segmented TurboFLASH and standard FLASH for cardiac imaging. The segmented TurboFLASH method exceeded the performance of the standard FLASH by allowing the capture of high-resolution cine images in one breath hold. In terms of image quality, the standard cine showed overall good quality but it had arrhythmia-induced ghosting artifact which obscured cardiac details. The single excitation scans showed respiratory ghosting while the two excitation scans reduced this artifact due to improved synchronization. The breath-hold cine technique which was the technique introduced in this paper showed no respiratory ghosting while the blurring of cardiac structures was less apparent than in standard cine studies. Moreover, the breath-hold requirement for this technique was 12-16 seconds corresponding to approximately 16 heartbeats in one phase of the respiratory cycle. The SNR was inferior to the standard cine technique but it was still adequate for image interpretation [10]. This showed great potential for the implementation of the single breath-hold segmented TurboFLASH in clinical practices.

The FISP (Fast Imaging with Steady-State Precession) method, similar to FLASH is another GRE-based method. It belongs to the SSFP family of techniques and derives its name from its Siemens implementation, as detailed in Table 1. The work carried out in 2001 studied the segmented TrueFISP technique in comparison with the standard FLASH. The results showed a normalized blood-myocardial CNR which was four times higher for true FISP compared to standard FLASH [22]. In another study, the parallel imaging method of SENSE was studied for real-time cardiac

imaging [25]. This study showed that SENSE can accelerate data acquisition greatly by using multiple RF coils. The parallel imaging methods generally cut scan times by up to two-thirds while also maintaining image quality. Therefore, they have been implemented with many MRI systems such as Compressed SENSE [39] which was explained in subsubsection 3.2.1. Despite the valuable advancements that were offered by the traditional methods, some limitations remained. For instance, reliance on breath-holding proved to be challenging for many patient populations, limiting accessibility and ease of use in clinical practice.

While the compressed sensing technique for MRI is considered a traditional technique in this review, it was developed in 2007 by Lustig et al. [15]. It was very cutting-edge at the time and it has laid the groundwork for recent DL and hybrid methods by introducing a sparsity-based reconstruction. It also addressed the challenge of dynamic imaging of time-varying objects such as the heart [13]. This challenge arises from the spatial and temporal sampling requirements of the Nyquist criterion. The traditional sampling of the k-space in an orderly fashion often fails to meet the spatial-temporal resolution requirements for dynamic imaging, leading to artifacts. With compressed sensing, random sampling of the k-space can overcome this issue and produce high-quality results with significantly fewer artifacts.

The recent methods, such as the ones that are based on DL or the hybrid approaches aimed to address all the previous challenges together. They aimed to enhance both speed and image quality, with the hopes of widening the clinical applicability and feasibility of CMRI scans. Some of these techniques introduce data-driven approaches that use neural networks to build on the strengths of the traditional methods; for example, CINENet [5]. There are also hybrid approaches such as Compressed SENSE, demonstrating the ability to achieve high-resolution imaging with faster speeds. All the progress in the area of MRI and CMRI helped the development of broader applicability in both clinical and research domains; since they enable real-time imaging while reducing patient discomfort.

5 Conclusion

This literature review has examined the evolution of MRI reconstruction techniques with a special focus on CMRI and cardiac cine MRI. The explained techniques ranged from traditional methods that focused on optimizing the hardware of the MRI system and the acquisition parameters to recent and emerging methods that use the power of ML and DL. These methods can be bridged with compressed sensing-based MRI, which revolutionized reconstruction by enabling the acquisition of high-quality images from undersampled data; thereby significantly reducing imaging time. The challenges remaining with compressed sensing such as the elimination of motion artifacts fully and achieving the robustness that is required in clinical applications were addressed by the recent advancements.

These advancements were followed by cutting-edge techniques such as DL-based methods and hybrid solutions. They demonstrated promising potential to make CMRI scans more feasible. However, the integration of these methods into clinical practice can be challenging due to the increased dimensionality with deep unrolled algorithms that rely on multiple iterations and the lack of training data availability [33]. Despite this, these methods hold great potential when the studies and the research that is poured into them are considered. Future advancements in MR hardware and data-driven approaches are expected to redefine CMRI, enabling more applicable real-time, high-resolution imaging with improved accuracy and efficiency. Continued research is essential to refine these methods and ensure their widespread clinical applicability, ultimately enhancing the diagnostic potential of CMRI.

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